More Punnett Squares

22

Recessive traits will be expressed if they are the only genes present. If they are linked to a dominant gene, then the dominant gene will be expressed. This is why Mendel's green peas only reappeared after a second generation. Recessive genes need time to come together again.

When recessive traits are crossed, the ratio of the recessive trait to the dominant trait in the offspring (F_1) is 1 : 3. This is illustrated by the Punnett square.

It's in the eyes!

Let's consider eye colour. In humans, brown eyes result from a dominant allele and blue eyes result from a recessive allele. Since each parent contributes an eye colour allele, a child might inherit one allele for blue eyes and one allele for brown eyes. To have blue eyes, offspring must have a double dose of recessive alleles, otherwise the dominant allele will be expressed.

1 Construct a Punnett square showing the possible outcomes of a cross between a brown-eyed (BB) father and a blue-eyed (bb) mother. (Remember: the female parent is shown on the side of the square and the male parent is shown on the upper part of the square.)

2 What will be the ratio of brown eyes to blue eyes?

_____ : _____

3 What will each offspring inherit in terms of eye colour alleles? _____

Consider what would happen if one of the offspring of the previous cross married a blue-eyed person. No-one can predict with certainty if any particular one of *their* children (the F_2 generation) will be blue-eyed or brown-eyed. However, the *probabilities* can be calculated.

4 Construct a Punnett square showing the possible outcomes of a cross between a female from the F_1 generation (Bb) and a blue-eyed male (bb).

5 What will be the ratio of brown eyes to blue eyes?

_____ : _____

6 The next generation would be called the _____ generation.

7 In a totally different family, two brown-eyed parents produced a blue-eyed child. What does this tell us about the genotypes of the parents?

_____

Time to think!

Gregor Mendel used the characteristics of peas to study inheritance. Most people might think that peas are boring and it might be more interesting (and useful!) to study humans.

8 Give one reason why Mendel didn't study inheritance in humans.

_____

© Peter Gribben, Wendy St Germain and Emerald City Books, 2001. This sheet may be photocopied for non-commercial classroom use.
© Blake Education 978 1 92561 060 4 Assignments in Genetics Book 1